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Optimizing Spreading Factor and Resources Allocation in LoRaWAN: A Hybrid Approach Using Reinforcement Learning and K-means Clustering

Mohamed Sofiane BATTA and Rahim KACIMI

***Abstract*—With the rapid expansion of Internet of Things (IoT) devices, current LPWAN technologies are increasingly susceptible to performance deterioration. LoRaWAN stands out as a leading LPWAN solution due to its ability to support long-range, low-power communication in massive IoT deployments. However, ensuring data reliability in dense and dynamic environments while optimizing transmission settings poses significant challenges, particularly in terms of interference mitigation and energy efficiency. To overcome these limitations, this paper proposes an innovative approach that integrates K-means clustering and leverages reinforcement learning to effectively manage gateway placement and configure transmission parameters, such as spreading factors and power levels, based on local signal conditions and environmental feedback. The model explicitly accounts for the capture effect and the imperfect orthogonality of spreading factors, allowing for more realistic and robust adaptation strategies without relying on the conventional centralized Adaptive Data Rate (ADR) mechanism. The approach is validated through a real-world experimental deployment. Furthermore, we highlighted the novelty of our approach by a detailed analysis of LoRaWAN spreading factor (SF) sensitivity thresholds, and evaluated the impact of gateway densification and SF reallocation on network performance. Both experimental and simulation outcomes show significant improvements in data extraction rate (DER) and energy efficiency, particularly in large-scale, congested network scenarios.**

***Index Terms*—IoT, LoRaWAN, Resources allocation, Spreading Factor, Reinforcement Learning, ADR**

I. Introduction

THE Internet of Things (IoT) is a transformative technology that has rapidly advanced in recent years. By enabling intelligent sensing and actuation, IoT allows a wide array of objects to interact with core networks for remote monitoring and administration through various systems. This technological revolution has sparked numerous applications, including smart homes, medicalcare systems and smart cities [1], [2]. As these applications continue to evolve, the number of IoT end devices (EDs) is projected to grow from 7 billion to 30 billion between 2018 and 2030 [3].

Fulfilling such scalability demands requires the development of smart, adaptive, and cost-efficient solutions to address the limitations of resource availability [4], Whereas traditional cellular networks aren't well-suited to meet IoT's transmission requirements, particularly in terms of long-range, low power communication [5], [6]. Consequently to these challenges, several Low-Power Wide-Area Network (LPWAN) protocols have evolved, e.g. LoRa, NB-IoT and LTE-M [5], [7], Among which LoRaWAN stands out due to its ease of deployment, low cost, and ability to deploy private networks without the need for external operators. It uses an open standard protocol and operates on unlicensed ISM frequency bands, allowing individual users to deploy private networks without regulatory constraints. These features make LoRaWAN particularly suited for scenarios with reliable operation in weak signal areas [7]. LoRaWAN has various physical layer parameters which are critical for optimizing network transmission performance, e.g. Spreading Factors (SFs), sub-channels, and transmission power (TP). For instance, a higher SF value increases the communication range and robustness, but at the cost of a lower data rate and longer Time on Air (ToA), which in turn raises the likelihood of transmission collisions. Balancing these trade-off is a critical for optimizing LoRaWAN deployments [8].

A. Motivation

LoRaWAN mainly implements an Adaptive Data Rate (ADR) mechanism aimed at optimizing network performences. However, the slow adaptation of this rule-based mechanism makes it inefficient for rapidly evolving network conditions [9]. Moreover, ADR relies on centralized control by the network server to computes transmission parameters, but frequent communication drains device energy and is limited by ISM band duty-cycle constraints. This highlights the need for autonomous, locally adaptive resource allocation to enhance efficiency and reduce server dependency.

While several autonomous resource management strategies have been proposed, most rely on mathematical models that approximate transmission metrics such as RSSI, often diverging from real-world conditions, particularly in large-scale deployments. Environmental factors like terrain, buildings, and weather further degrade communication quality, limiting scalability and performance [10]. Experimental results in [8] highlight these shortcomings and the need for more adaptive methods. In this context, involving reinforcement learning (RL) emerges as a promising solution, offering dynamic resource allocation by enabling devices to learn and optimize their settings based on real-time environmental feedback.

B. Novelty claims

Building on these ideas, we propose an autonomous resource allocation scheme based on RL. Our solution combines

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two learning techniques, K-means for dynamic gateway placement and Q-learning for transmission parameter optimization (SF and TP). The K-means model identifies optimal gateway positions to enhance coverage, while the Q-learning model allows ED's to autonomously adjust their settings, considering the capture effect and imperfect SF orthogonality. The approach is designed for high-traffic scenarios and aims to improve the Quality of Service (QoS), reduce energy consumption, limit inter-SF interference, and maximize the data extraction rate (DER).

To better underline the major contributions and the novelty of this work, we dedicate this paragraph to enumerate the threads of this paper:

- LoRaWAN SF threshold experimental study: in section VI-B we consolidate the novelty of our approach by a detailed analysis of the real LoRaWAN SF sensitivity thresholds, which, to the best of our knowledge, is not discussed in previous literature studies.
- In section IV we propose a novel framework that combine unsupervised learning to predict optimal gateways placement, and exploits RL to enables the ED's to adequately configure their transmission parameters while considering the capture effect and the imperfect orthogonality of SFs using an autonomous adaptive strategies.
- Experiments: in section VI the novelty of our work is endorsed by addressing a real-world autonomous protoype deployed and tested over standard LoRaWAN devices. The implementation of our experiments is publicly available [11] for further use and potential analysis by the LoRaWAN research community.

The remainder of the paper is ordered as follows: Section II reviews related works. Section III provides an overview of LoRa and LoRaWAN. Section IV presents our proposed approach in detail. Section V-B describes the interferences system model. Section VI presents the experimental setup, simulations, and result analysis. Additionally, an experimental deployment is described in this section to provide valuable insights and demonstrate the performance of the proposed approach in a real LoRaWAN deployment. Finally, Section VII provides concluding remarks and future research directions.

II. Related works

Numerous studies have examined LoRaWAN performance, addressing major challenges such as scalability, interference, and network limitations [4], [5], [7]. Beyond the communication technology itself, network performance largely depends on how wireless resources are managed [12]. To overcome these challenges, researchers have proposed various solutions focused on optimizing SFs, energy consumption, and channel allocation to improve the DER and extend device lifetimes [13]. This section provides a brief overview of these resource allocation strategies, categorized into conventional, machine learning-based, and prototype-based approaches. The key contributions of the reviewed studies are summarized in Table I.

A. Conventional proposals

One of the pioneering conventional approaches in the literature addressing SF allocation to enhance LoRaWAN network performance is presented in EXplora [14]. This work aims to increase the DER by employing a fair SF allocation by factoring in communication Signal Strength (RSSI) and airtime. It uses a sequential waterfilling approach to ensure low Time on Air and maintain fairness across channels. Building on this strategie, several variants have emerged. For instance, the approach in [15] addresses radio resource allocation in multi-tenant LoRaWAN networks using network slicing to support heterogeneous applications with different QoS requirements. The authors propose to assigns channels and SFs to EDs in a multi-operator environment, aiming to maximize normalized throughput while ensuring fairness across network operators. The EXplora-Capture (EXplora-C) [16] refines EXplora-TS by balancing the ToA of uplink packets while considering the capture effect in both single and multi gateway environments. The study in [17] formulates an optimization problem to minimize collisions by leveraging the capture effect for optimal SF distribution. Their algorithm dynamically adjusts SNR thresholds to enhance throughput.

Although conventional mechanisms deliver promising results, they exhibit limitations in adapting to dynamic environmental changes. This has led to the exploration of RL-based methods to maximize network performance.

B. RL-based approaches

Many researchers have applied RL techniques to address optimal resource allocation in LoRaWAN networks [3], [13]. LoRaDRL [18] introduces a deep RL method based on Deep Q-Network, designed for intelligent multi-channel resource allocation in dense networks. LoRaDRL aims to learn the optimal policy for allocating resources to mobile EDs by considering channel conditions and network traffic loads. Similarly, A hybrid solution combining TinyML and a CNN-LSTM model on the network server has been proposed in [19] to optimize resource allocation for mobile applications. This approach enables the prediction SFs and TPs by exploiting spatial and temporal patterns in real-time data. Their model, mounted at the gateway, minimize collisions rate and optimize the network energy efficiency. However, simulations were conducted within small network area, which may not generalize to larger deployments. An enhanced Q-learning STEPS (Score Table-based Evaluation and Parameters Surfing) approach is presented [20]. The approach applies a score table to optimize SF by adapting transmission parameters based on the success probability of packet reception and aims to reduce the excess of energy consumption.

Although ML-based methods outperform conventional ADR schemes, they often depend on centralized architectures requiring frequent communication with the network server, which impacts scalability and collision risks. Additionally, ISM duty cycle limitations present further constraints. These challenges underline the need for autonomous resource management strategies better suited to large-scale real-world deployments.

C. Experiment based approaches

This section highlights studies that leverage real-world prototypes, so far, only a few valuable subset of studies

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have validated their solutions in actual deployment environments. LoRa-MAB [21] addresses scalability limitations of centralized RL by adopting a distributed Multi-Armed Bandit approach, allowing devices to locally adapt SFs based on ACK feedback. While it improves DER, it overlooks energy efficiency and relies on confirmed mode, introducing extra ACK traffic that increases interference and energy consumption, especially in dense networks. In [8], the authors proposed L3SFA, a centralized load-shifting technique that dynamically adjusts SF allocation based on traffic load and ED density. By assigning the smallest SF compatible with each ED's RSSI, L3SFA reduces ToA and balances communication load across SFs. Simulations, validated with real deployment data, demonstrate significant energy savings without degrading DER compared to standard ADR.

While experimental studies provide valuable insights into LoRaWAN resource allocation, most existing approaches still lack validation through real-world prototypes. In contrast, beyond our proposed RL-based allocation scheme, the novelty of this work is endorsed by addressing a real-world prototype deployed and tested on standard LoRaWAN devices. To promote reproducibility and foster further research, our implementation is publicly available [11].

III. LoRa and LoRaWAN Review

The LoRa network architecture is built on two key components: LoRa and LoRaWAN. LoRa, developed by Semtech, is a proprietary physical layer that employs chirp spread spectrum (CSS) modulation to enhance signal robustness against interference [3]. This enables LoRa to achieve a power budget exceeding 150 dB in the 868 MHz band and decode signals up to 20 dB below the noise floor, supporting long-range communication [17]. LoRa's performance depends on several configurable parameters, including the SF, TP, bandwidth (BW), coding rate (CR), and carrier frequency, each representing a trade-off between energy efficiency and communication robustness [4]. The quasi-orthogonality of SFs allows concurrent transmissions, enhancing scalability [7]. However, selecting the optimal SF involves balancing range, data rate, and energy consumption. Higher SFs extend range but increase energy use and lower throughput, while lower SFs reduce ToA and interference but offer less robustness. LoRaWAN parameters are summarized in Table III, and Table IV illustrates the relation between SFs, data rate, sensitivity, and SNR thresholds. The nominal bit rate as a function of SF is defined in equation 1.

$$DR = \frac{SF \times CR}{2^{SF}/BW} \tag{1}$$

LoRaWAN is the open upper-layer protocol defined by the LoRa Alliance atop the LoRa physical layer. It employs a star topology connecting EDs, gateways, a network server (NS), and an application server (AS), as shown in Fig. 1. The NS oversees network access, message integrity, and parameter tuning (SFs and TPs) via the ADR mechanism, while the AS ensures data encryption and supports end-user applications.

LoRaWAN defines three classes of end devices (A, B, and C), each balancing energy efficiency and downlink latency differently (Fig. 2). Class A, the most energy-efficient, opens two short receive windows after each uplink, suiting applications where devices that mainly send data. Class B introduces scheduled receive slots for more regular downlinks, while Class C continuously listens for messages, offering minimal latency at the cost of higher energy consumption [13].

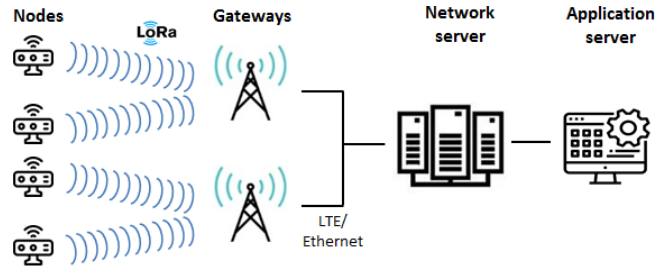

Fig. 1. LoRaWAN network architecture

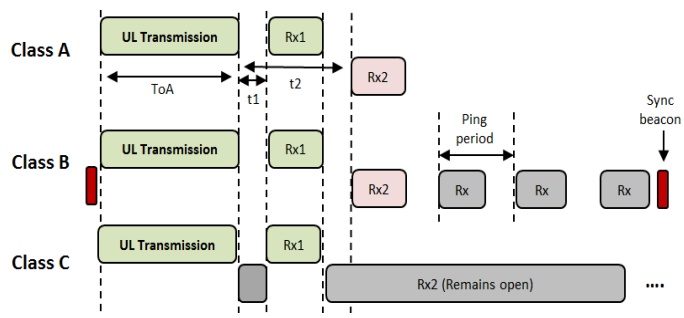

Fig. 2. LoRaWAN end nodes classes

IV. The proposed approach: DSF2R

To clarify the comprehension of our approach, we divide it into two phases. Firstly, we introduce a K-means clustering method designed to predict optimal gateway placements based on the network topology to ensures strategic coverage and reduces transmission distances. Secondly, we present a Q-learning model that enables EDs to autonomously adjust their SF and TP. This adaptive mechanism aims at reducing energy consumption, and minimizing intra-SF interference, and eventually increasing the network DER.

A. Phase 1: GW Location Optimization

The proposed approach forms clusters represented by centroids, following the principle of non-overlapping Voronoi diagrams [25]. Given the set of EDs coordinates $S = \{n_1, \ldots, n_x\}$, the gateway placement algorithm partitions S into k non-overlapping clusters $C = \{C_1, .., C_k\}$ such that:

$$S = \bigcup_{i=1}^{k} C_i \quad \text{and} \quad \forall i \neq j : C_i \cap C_j = \emptyset$$

The algorithm uses a k-means clustering approach [3] to finds proper partition of nodes in a way that minimizes the within-cluster sum of squares ($WCSS$) depicted in equation 2.

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TABLE I
SF ASSIGNMENT APPROACHES FOR LORAWAN NETWORKS PROPERTIES COMPARISON

Algorithm	Distr. ADR	RL Category	Dynamic GW	Features	Enhancement Benchmarks
EXplora-AT, EXplora-SF [14]	X	X	X	RSSI, ToA	DER
Lodhi [19]	✓	✓	X	ToA, Data payload	DER, Energy consumption, mobility
Romero [15]	X	X	X	RSSI, QoS	DER, throughput, latency
EXplora-TS, EXplora-KM [22]	X	X	X	RSSI, ToA, Data payload, traffic load	DER, channels load balancing
EXplora-C [16]	X	X	✓	ToA, Data payload, traffic load, capture effect	DER
Loubany et al. [17]	X	X	✓	SNR, capture effect	DER, intra-interference
LoRaDRL [18]	X	✓	X	RSSI, ToA, traffic load, mobility	DER, Energy consumption
MARL [23]	X	✓	X	RSSI, Energy, packets collisions	DER, Energy consumption, interferences
STEPS [24]	✓	✓	X	Success reception probability	DER, Energy consumption
L3FSA [8]	X	X	X	RSSI, T_x power	DER, Energy consumption
LoRa-MAB [21]	✓	✓	X	ACK	DER
Proposed approach	✓	✓	✓	SNR, T_x power, net. topology	DER, Energy consumption, Interferences

$$WCSS = \min_{\forall c_i \in C} \sum_{i=1}^{k} \sum_{n \in C_i} \left\| n_i^{(C_i)} - c_i \right\|^2 \tag{2}$$

Where $n_i^{(C_i)}$ is the set of ED's assigned to cluster i and c_i is the centroid of cluster i. Lower $WCSS$ values indicate more cohesive clusters where ED's within the same cluster are as close as possible to their centroid. The K-means algorithm incorporates the Okumura–Hata urban propagation model [17], used to estimate the RSSI for each ED–centroid pair. This model demonstrates a balanced approach for path loss and aligns closely with results observed in experimental setups. The environment parameters used in this model are highlighted in tableIII. The number of gateways K to be deployed, is a user-defined parameter that depends on the network size, density, and deployment constraints. The centroid of the configuration yielding the minimum within-cluster RSSI-based variance $WCSS$ is selected as the gateway location. The workflow of our dynamic GW's placement process is illustrated in Figure 3.

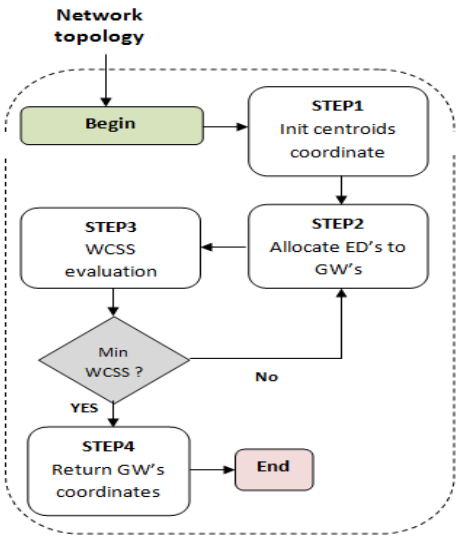

Fig. 3. Gateway location algorithm process

B. Phase 2: SF and Tp Allocation Using RL

In this section, we present an embedded Q-learning model that allows EDs to autonomously adjust their SF and TP. The learning framework is formulated as a Markov Decision Process (MDP) $M = \{S, A, R, P\}$, where S denotes the state space, A the action space, R the reward function, and P the state transition probability function. The resource allocation learning process is illustrated in Figure 4.

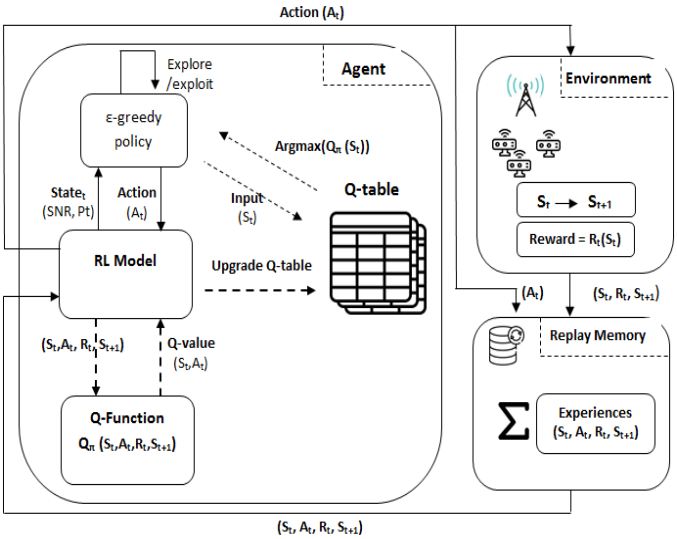

Fig. 4. $DSF2R$ learning framework

During training, the model associates each action-value function $Q^{\pi}(S, A) = E_{\pi}\{R_t | S_t = s,\ A_t = a\}$ with a Q-value that represents the expected quality of taking action A in state S. Where, $\mathbb{E}_{\pi}$ denotes the expected return when following policy π. Specifically, the Q-value for a state-action pair (S_t, A_t) in our model is computed as shown in equation 3.

$$\begin{aligned} &Q(S_t, A_t) = Q(S_t, A_t) + \\ &\alpha \times [R_t + \gamma \times \max_{\pi} Q(S_{t+1},\ A_{t+1}) - Q(S_t, A_t)] \end{aligned} \tag{3}$$

The objectives of the model are governed by the design of the reward function and the definition of the state space. Upon receiving a downlink message from the gateway, the new state is determined based on the measured SNR and the current TP of the ED. The reward is then computed according to Equation 4, and the corresponding Q-value is updated using Equation 3. After updating the Q-table, new SF and TP values are selected following an ε-greedy strategy as defined in Equation

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6. The training process is summarized in the pseudo-code presented in Algorithm 1. In the following, we detail the specific parameters that define our learning model.

Algorithm 1 Training phase

Initialization
Discrete SNR space to 55 values
Initialize Q-table with arbitrary values
Initialize buffer (batch memory) β_m to capacity $|S|$
$\beta_m = |S = \{n_1, .., n_x\}|$

For $Episode\ t = 1, M$ **Do**
Initialize network environment (nodes states)
For each ED_i **Do**
$State_t^i = (SNR_t^i, Pt_t^i)$
Use the $\epsilon - greedy$ policy to chose action A_{t+1} at state S_t
$A_{t+1} = \begin{cases} \max_\pi Q(S_t) & \text{if } n < \epsilon\ (exploit) \\ \text{Random action} & \text{otherwise } (explor) \end{cases}$
Execute action A_{t+1} in the environment
Get reward R_t using equation 4; and next state S_{t+1}
Add experience (S_t, A_t, R_t, S_{t+1}) into buffer β_m
$S_t \leftarrow S_{t+1}$
End for
Fit replay buffer β_m to the target Q-function (equation 3)
Compute Q-values using equation 3 and Update Q-table
Empty Replay buffer $\beta_m = \emptyset$
Update ϵ value: $\epsilon_t = \min(\epsilon_{min}, \epsilon_t - \frac{1}{N})$
$t \leftarrow t + 1$
End for

*1) **Reward Function**:* We designed the reward function to encourage energy-efficient communication while ensuring reliable connectivity. Specifically, it relies on two key metrics:

- **The Spreading Factor (SF)** which strongly influences energy consumption and time-on-air.
- **The SNR margin** defined as the difference between the measured SNR and the sensitivity threshold associated with the selected SF.

When the communication link is reliable (i.e. positive SNR margin), the reward combines these two aspects as defined in equation 4.

$$R_t = \frac{\frac{1}{SF} + SNR_{margin}}{SF \times SNR_{margin}} \tag{4}$$

The term $\frac{1}{SF}$ encourages the use of lower SF values, thus promoting energy savings, while SNR_{margin} rewards configurations that maintain a strong and robust link.

When the SNR margin becomes negative, the model applies penalty using equation 5, which explicitly discourages configurations that either degrade link reliability or increase energy consumption without benefit. As a result, the agent progressively learns to balance link quality and energy usage through autonomous and environment aware adaptations.

$$R_t = SF \times SNR_{margin} \tag{5}$$

*2) **States and Actions**:* In our model, states are defined by the pair (SNR, TP), while actions correspond to the pair (SF, TP), with permissible values given in Table III. This enables each ED to dynamically adjust its configuration based on the current conditions. An $\epsilon - greedy$ strategy (equation 6) is used to balance exploration and exploitation during training. The SNR range, observed between 6 and -20 dB, is discretized into 50 levels for finer state representation. Combined with 6 possible TP values, which results in 50×6 distinct states, each representing a specific environmental condition based on the SNR level. Assuming symmetric links between EDs and GWs, the same Q-table can be shared across EDs, supporting an adaptive plug-and-play model that ensures autonomous operation without degrading algorithm performance.

$$A_{t+1} = \begin{cases} \max_\pi Q(S_t) & if\ n < \epsilon \\ Random\ action & otherwise \end{cases} \tag{6}$$

V. System model

We consider a LoRaWAN network comprising N EDs and $k \in \{1, 2, 3\}$ GW's, dynamically positioned using the gateway placement algorithm IV-A to assess the impact of network densification. Class A EDs transmit unconfirmed messages over symmetric links, following a Poisson process with uniform arrival rate λ packets per second. Let N be the total number of EDs, N_c the number of covered EDs, and N_i the number on devices assigned to SF_i. This setup provides a realistic framework to evaluate network capacity, coverage, and performance under different gateway densities. Figure 5 illustrates a sample node deployment with $N = 600$ ED's distributed over a square area of 4 km per side and served by three dynamic gateways. Although, we limited the system model with three GW's to provide a controlled and interpretable evaluation of densification effects. The results drawn from the experimental scenarios are not restricted to small scale deployments. The observed trends are general and remain applicable to larger networks with a higher number of gateways.

Table II enumerates a list of abbreviations and some variables used in this paper.

A. DER Computation

A key parameter for evaluating network efficiency and reliability is the Data Extraction Rate (DER). A higher DER reflects better network performance, indicating that a majority of transmitted packets are successfully received by the GW. The network DER is influenced by several factors such as interference, SF allocation, link SNR, and GW placement. As LoRaWAN relay on Aloha access scheme, the channel traffic load, or the average number of packets transmitted per time ToA_i for each SF_i is given by equation 7. Where N_i denotes the total number of nodes that have been assigned to SF_i, λ is the message generation rate and C is the number of uplink channels.

$$Load_i = \frac{\lambda \times N_i \times ToA_i}{C} \tag{7}$$

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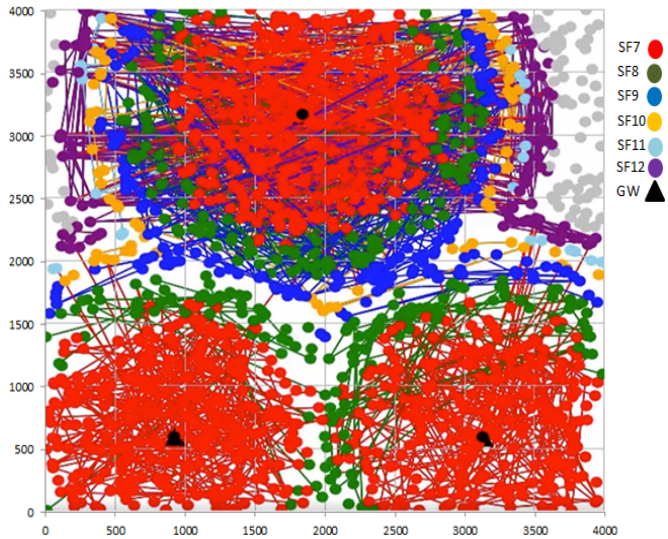

Fig. 5. SF Distribution for 600 end-devices in a cell of 4000 meters grid served by three dynamic GW's

TABLE II
ABBREVIATIONS AND VARIABLES

Abbreviations	Description
SF	Spreading Factor
TP	transmission power
ToA	Time on Air
ADR	Adaptive Data Rate
DER	Data Extraction Rate
RSSI	Received Signal Strength Indicator
SNR	Signal-to-noise ratio
QoS	Quality of Service
RL	Reinforcement Learning
GWs	Gateways
EDs	End devices
CR	Coding rate
BW	Bandwidth
K	Number of deployed Gateways
S_t	Environement State at time t
A_t	Learning agent action at time t
N	Total number of devices
N_c	Number of covered devices
N_i	Number on devices assigned to SF_i
$Load_i$	Channels traffic load for each SF_i
DER_i	Data extraction rate achieved for SF_i
λ	Packet generation rate
C	Number of uplink channels

Consequently, according to the Poisson distribution, the DER_i achieved for SF_i, given a channels load $Load_i$ is derived as in equation 8.

$$DER_i = \epsilon^{-2\times Load_i} \tag{8}$$

The optimization problem for SF allocation involves determining the optimal distribution of nodes across each SF to maximize network performance. This optimization problem can thus be formulated as shown in equation 9.

$$\begin{aligned} &\text{Maximize} && \tfrac{1}{N}\textstyle\sum_{i=7}^{12} \mathrm{DER}_i \times N_i \\ &\text{Where} && \textstyle\sum_{i=7}^{12} N_i = N \end{aligned} \tag{9}$$

B. Interferences Model

Although concurrent LoRa transmissions using different SFs can theoretically be decoded by the gateway, practical limitations arise due to the imperfect orthogonality of SFs, leading to mutual interference and increased error rates when signal strengths vary significantly. However, when two signals overlap, data loss is not systematic, indeed, if the signal strength difference remains below a certain threshold, one or both signals may still be decoded successfully [8]. The required signal isolation for successful reception is captured by the SIR matrix (Table 12), which specifies the minimum SIR between SF pairs. Achieving these levels typically requires mechanisms such as TP control. In this context, interference is generally classified as either intra-SF or inter-SF collisions.

*1) **Intra-SF interferences and Capture Effect**:* Intra-SF interference arises when two frames are transmitted simultaneously on the same channel with identical SF, i.e., $SF^i = SF^j = x$. Such collisions do not always lead to data loss thanks to the capture effect. Indeed, if the $RSSI$ difference between the two frames exceeds 6 dB (as shown in equation 10), the frame with the stronger signal can still be decoded. Otherwise, both frames are lost [21].

$$\|RSSI^i_{sf} - RSSI^j_{sf'}\| \leq 6 \tag{10}$$

*2) **Inter-SF interference**:* Because of the partial non-orthogonality of SFs, simultaneous transmissions from frames i and j using different SFs can interfere with each other. A frame is successfully decoded if the signal strengths meet the SIR thresholds defined in the SIR matrix (Table 12) [8]. Otherwise, when the interference exceeds these limits (equation 11), only the stronger signal frame i can be received.

$$RSSI^i_{sf} - RSSI^j_{sf'} \leq SIR_{sf,sf'} \tag{11}$$

$$SIR[dbm] = \begin{bmatrix} 6 & -8 & -9 & -9 & -9 & -9 \\ -11 & 6 & -11 & -12 & -13 & -13 \\ -15 & -13 & 6 & -13 & -14 & -15 \\ -19 & -18 & -17 & 6 & -17 & -18 \\ -22 & -22 & -21 & -20 & 6 & -20 \\ -25 & -25 & -25 & -24 & -23 & 6 \end{bmatrix} \tag{12}$$

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Model training Settings

The simulation and training setup is built upon a modified version of the Python-based LoRaSim framework [26], tailored to incorporate the proposed RL approach for dynamic parameter allocation. Key enhancements include: (a) realistic modeling of both intra- and inter-SF interference; (b) incorporating a fading model to better reflect environmental signal propagation effects; (c) uniform frequency allocation to handle channel congestion; (d) integration of a learning model for dynamic optimization of transmission parameters; and (e) implementation of a LoRaWAN energy consumption model based on established references [27], [28], allowing precise evaluation of energy efficiency.

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The Q-learning model is trained on a network of 1000 EDs uniformly deployed over a 4000×4000 m^2 area. We used a real-world dataset of LoRaWAN messages, collected from deployments on the university campus and across various zones in Toulouse. The dataset messages contains EDs metadata describing the channel state (RSSI, SNR) and transmission parameters (SF, BW, CR, etc). These real-world values were used as simulation inputs to train our model under realistic radio conditions. The total training duration is 2 hours using Colab NVIDIA Tesla K80 GPU (12GB VRAM) during which the model runs 100k episodes of training. Table III summarizes the key simulation and training parameters. The implementation of our prototype is available on GitHub [11].

To thoroughly evaluate the performance of the proposed approach, we conducted an extensive set of experimental and simulation scenarios, presented in the following subsections. For statistical robustness, each simulation scenario was executed 10 times with different random deployments of EDs. Final results are reported as the average performance across all runs.

TABLE III
EXPERIMENTAL SETTING AND TRAINING PARAMETERS

Parameter	Value
Network size	$4000 \times 4000\ m^2$
Number of nodes, N	1000
TP	$[6, 8, 10, 12, 14]$
SF	$[7, 8, 9, 10, 11, 12]$
Channels used (scenario 1)	$3 : [868.1, 868.3, 868.5]$
Channels used (scenarios 2, 3)	$8 : [868.1, 868.3, 868.5, ..., 867.9]$
SNR Thresholds and Sensitivity	Table IV
Inter-SF and Intra-SF Interferences	Matrix 12
packet generation period $1/\lambda$:	100, 300, 600 (seconds)
BW, CR	125 kHz, 4/5
Payload	20 bytes
Path loss model	Okumura–Hata Urban [17]
Discount factor γ, Learning rate α	0.9, 0.1
ED's / GW's Height	1 meter, 15 meters

B. Experiment: SFs Sensitivity Threshold Characterization

To gain a deeper understanding of how SFs sensitivity thresholds behave in real-world conditions according to the signal quality between the end modules and the LoRa GW, we conducted a dedicated analysis based on the data collected during our experiments (Section VI-A). While many existing works that focus on optimizing LoRaWAN transmission rely on theoretical SFs sensitivity thresholds commonly cited in the literature. To the best of our knowledge, no comprehensive empirical study validating these thresholds using real hardware and environmental conditions. The objective of this experiment is twofold:

- To experimentally characterize the effective sensitivity thresholds for each SF under realistic deployment conditions.
- To reinforce the modeling of critical phenomena such as the capture effect and the imperfect orthogonality of SFs, both of which significantly influence network performance in dense scenarios.

In this regard, we measured the DER across varying SNR levels for each SF, using actual LoRaWAN devices under realistic field conditions. In Figures 6, we illustrate the evolution of DER with respect to SNR, where a reception margin of 15/50 successfully received packets was considered to determine empirical sensitivity boundaries respectively. Corresponding numerical values are summarized in Table IV.

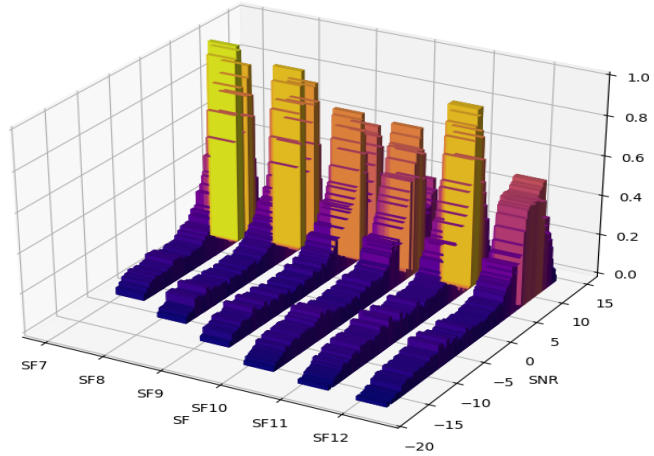

Fig. 6. Spreading Factors Sensitivity Threshold according to signal SNR (The vertical axis represents the DER ratio)

TABLE IV
DATA RATE, SENSITIVITY AND SNR THRESHOLDS OF DIFFERENT SFS FOR 868 MHZ BAND (BANDWIDTH 125 KHZ, CODING RATE 4/5)

SF	Data rate [kb/s]	Sensitivity[dBm]	Required SNR [dB]		
			Conv.	1/50	15/50
7	5.458	-126.5	-7.5	-11.5	-9
8	3.125	-127.25	-10	-14	-11.5
9	1.757	-131.25	-12.5	-15.5	-14.5
10	0.976	-132.75	-15	-18	-16.5
11	0.537	-133.25	-17.5	-19	-17.75
12	0.293	-134.5	-20	-20.25	-18

Our findings indicate a noticeable discrepancy between the theoretical thresholds (provided in [13]) and those derived from real-world measurements. Specifically, real thresholds tend to deviate from the conventional sensitivity values commonly used for balancing SF distribution in simulation-based studies. These inaccuracies can lead to suboptimal allocation strategies, especially under high SF usage, by overestimating reception capabilities, which may in turn increase the risk of interference and degrade the effectiveness of capture effect exploitation. Therefore, in the remainder of this work, we adopt the experimentally derived SF sensitivity thresholds measured under typical LoRaWAN deployment conditions, as they offer more accurate basis for optimizing resources allocation while mitigating inter- and intra-SF interference.

C. Scenario 1: Experimental Evaluation

This first scenario presents an experimental evaluation conducted using a real-world testbed deployed on the Toulouse University campus. The deployment, illustrated in Fig. 8,

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covers an environment with a radius of approximately 1.5km. This testbed comprises three gateways and several EDs based on Raspberry Pi 4 Model B, each equipped with a Microchip RN2483 LoRa transceiver, as shown in Fig.7. This setup allows realistic validation of our proposed approach under practical conditions. For the backend infrastructure, we employed Mosquitto [29], an open source message broker that implements the MQTT protocol chosen for its lightweight architecture and modular design.

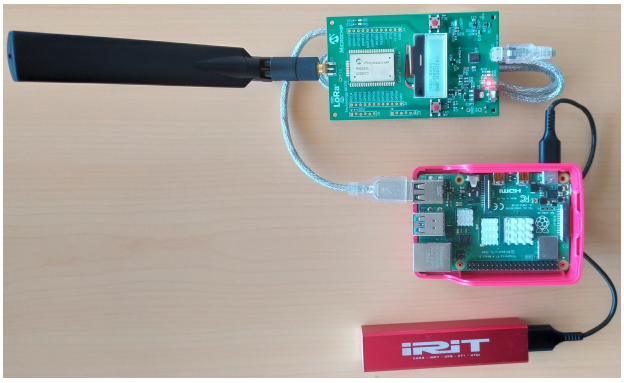

Fig. 7. Experimentation Setup with LoRa microchip-based RN2483 and Raspberry Pi4 model B ED

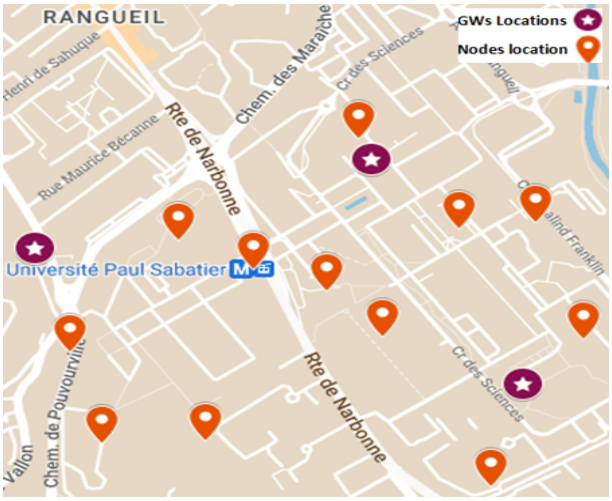

Fig. 8. Experimentation deployment with 3 GWs and 12 RN2483 EDs locations (within Toulouse University campus)

The primary objective of this experiment is to evaluate the performance of the proposed learning-based approach in comparison to the standard ADR mechanism, with a particular emphasis on energy consumption as a key efficiency metric. For this experiment scenario, we only consider the three mandatory default channels in the EU868 band with a packet generation interval of $p = 6s$. Using a short transmission interval combined with a reduced number of available channels increases channel occupation and replicates high-density conditions, thereby inducing network congestion. This setup choice is motivated by the goal of the experiment to overload the network and evaluates the robustness of the proposed approach under overloaded conditions. This is possible either by using numerous end-devices, which is not practical in this instance due to the lack of hardware platforms, or by setting the packet generation period per node λ to the minimum and increasing the packet payload. Indeed, generating heavy traffic using a large number of physical end devices is not feasible due to hardware limitations. Overloading the network can instead be emulated by reducing the channel diversity and imposing a high traffic generation rate.

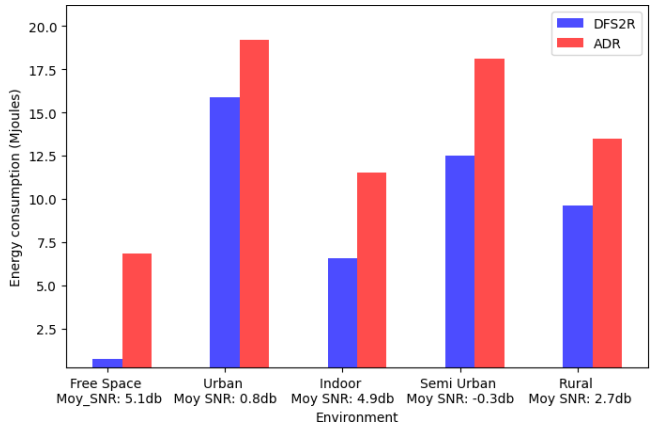

Fig. 9. Comparison of energy consumption of DFS2R vs ADR according to different environments.

TABLE V
STATISTICS RESULTS COMPARISON THE PROPOSED RL APPROACH VERSUS THE ADR STRATEGY (WITH 500 MESSAGES)

	Energy consumption (Mjoules)				
	Free Space	Urban	Indoor	Semi Urban	Rural
DSF2R	0.72	15.9	6.54	12.5	9.62
ADR	6.84	19.2	11.5	18.1	13.49

Fig. 9 and Table V present the average energy consumption results obtained across different environments, based on the transmission of 500 uplink messages. As illustrated in Fig. 9, the proposed approach outperforms the conventional ADR mechanism, achieving an average energy savings of 34.6%. This improvement can be attributed to the fact that our model is trained on a dataset encompassing different environmental conditions, enabling it to make more informed and adaptive parameter allocation decisions. In contrast, the standard ADR algorithm bases its adjustments solely on a limited history of recent successful transmissions, making it less responsive to dynamic and heterogeneous radio environments.

D. Scenario 2: Simulation Evaluation

In addition to the experimental validation, multiple simulations were complimentarily conducted under varying packet traffic densities ($\lambda \in [1/100,\ 1/300,\ 1/600]$). This range was selected to enable a comprehensive evaluation of the network's scalability and its responsiveness under different traffic loads. For this purpose, we used LoRaSim [26] to simulate realistic network communication scenarios. The proposed approach

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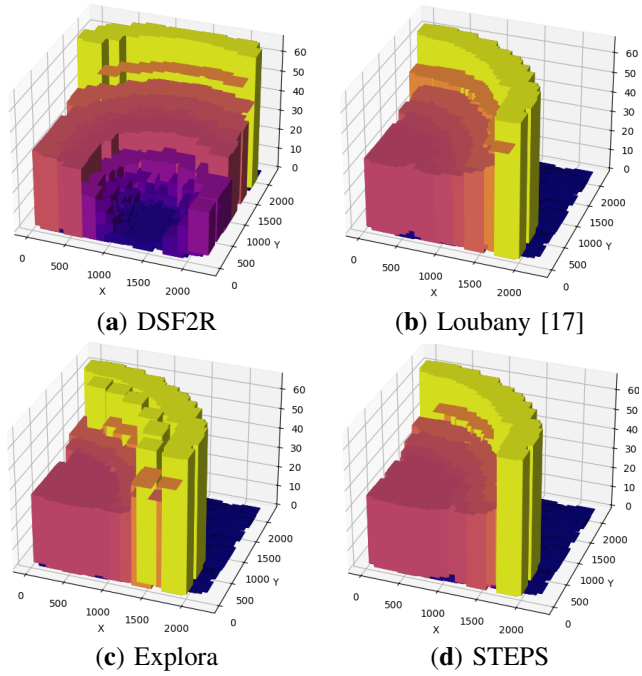
was evaluated against several approaches, including STEPS [20], Loubany [17], and Explora [16]. Energy consumption was chosen as the primary performance metric to quantify the efficiency gains offered by our model.

To better illustrate the impact of SF allocation strategies on network behavior, Fig. 10 displays the SF distribution and corresponding energy consumption for 700 EDs deployed within a 3 km^2 area. This figure maps the energy consumed by each node under the different approaches. As shown in Fig. 10(a), the proposed learning-based model demonstrates efficient utilization of radio resources and adapts well to the network topology, particularly optimizing SF and TP for EDs in proximity to the gateway. This optimization results in a more balanced energy distribution across the network. A deeper analysis of the total energy consumption, as depicted in Fig. 11, confirms that our approach achieves an average energy reduction of 32.3% compared to similar approaches. These simulation results align with the experimental results obtained in scenario 1 VI-C, thereby validating the effectiveness of our model in both real and simulated environments.



(**a**) DSF2R (**b**) Loubany [17]

(**c**) Explora (**d**) STEPS

Fig. 10. Distribution of energy consumptions. The vertical axis represents the energy consumption for each LoRaWAN node (MJoules).

E. Scenario 3: Gw's Densification Analysis

In this scenario, we evaluate the impact of GW's densification on the performance of the proposed model. To this end, we consider three deployment configurations involving one, two, and three gateways, with their placement determined using the algorithm described in Section 3.

Figure 12 illustrates the evolution of the DER under varying ED's densities and a high traffic load condition ($\lambda = 1/100$). The results reveal that increasing the number of GWs significantly enhances DER across all ED density levels. Specifically, the average DER improves from 39.3% in the single-GW scenario to 54% with three GWs, demonstrating the benefits of densification in improving reliability. In the single-GW setup, DER performance deteriorates sharply as the number of EDs increases, falling from 87.5% with 100 EDs to just 12.5% with 1000 EDs. This degradation is primarily due to the increase of intra SF interference, as shown in Fig. 14, resulting from channel contention and insufficient spatial coverage in dense deployments. Conversely, multi-gateway deployments lead to a noticeable stabilization of DER within the $[30\% - 50\%]$ range, even under high ED densities. This improvement is attributed to better spatial reuse, reduced interference levels, and enhanced coverage. As a result, the network is more capable of sustaining higher loads without compromising communication reliability. This observation underscores the effectiveness of multi-gateway configurations in managing high-density networks and maintaining data reliability in LoRaWAN systems under demanding traffic conditions.

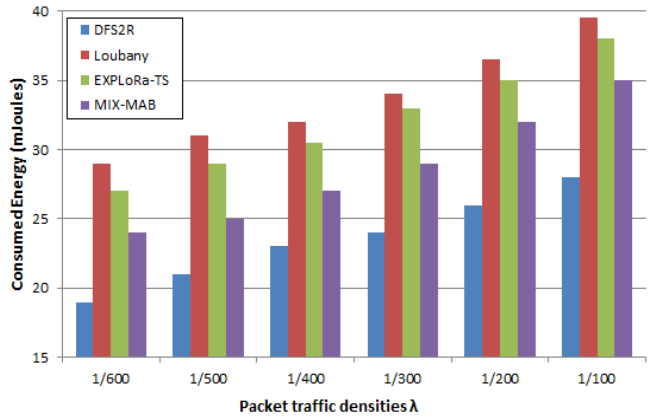


Fig. 11. Average energy consumption according to transmission traffic density

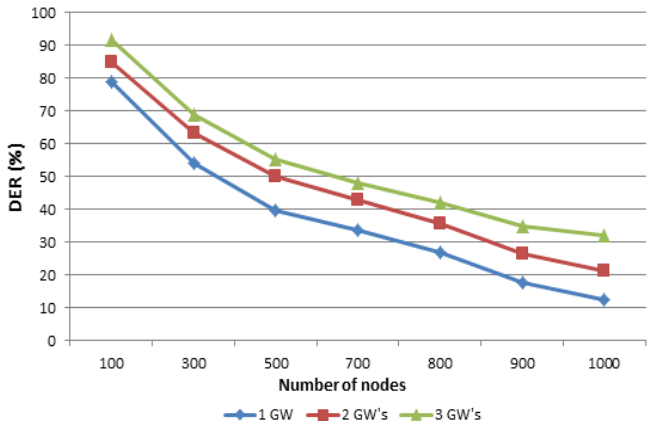


Fig. 12. Average DER ratio across three gateway scenarios under varying ED's densitie ($\lambda = 1/100$))

However, gateway densification also significantly affects the distribution of SFs, as illustrated in Fig. 13. Notably, the proportion of EDs assigned to $SF7$ increases substantially with the number of GW's, rising from 21.8% in the single-GW scenario to 50.1% in the three-GW configuration. Meanwhile, the proportion of nodes allocated to other SFs remains con-

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sistently below 14% across all deployments. This increased concentration at $SF7$ stems from the improved signal quality and higher RSSI values enabled by better spatial coverage in multi-gateway topologies. While favorable in terms of energy efficiency and latency, this uneven distribution introduces a heigh risk of collisions, particularly in areas with high node density. As depicted in Fig. 14, the average intra-SF interference calculated per deployment scenario exhibits a notable decline as the number of gateways increases. Specifically, the average intra-SF interference rate drops from 58.7% with a single GW to 40.5% with three GWs.

Overall, these results highlight a critical trade-off: while gateway densification enhances coverage, link quality and promotes lower SF usage and reduce ToA, it simultaneously leads to a concentration of EDs on lower SFs, since low SFs become sufficient to maintain reception thresholds, which may rise congestion and intra-SF interferences. The proposed approach redistributes devices across multiple feasible SFs to mitigate this trade-off and balance transmission efficiency and collision mitigation in dense, multi-gateway LoRaWAN networks.

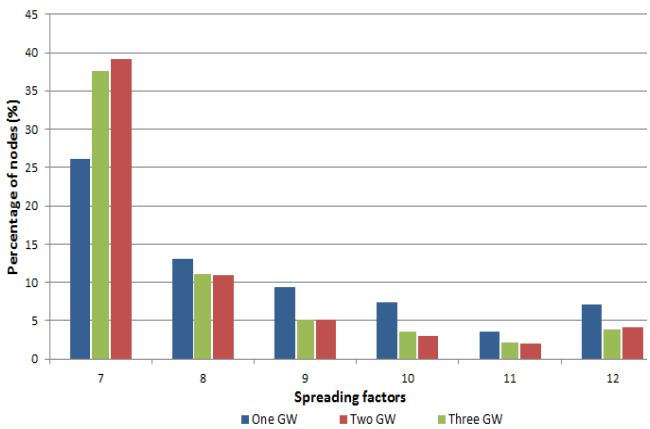

Fig. 13. Nodes distribution at each SF under different gateway scenarios (700 ED's with $\lambda = 1/100$))

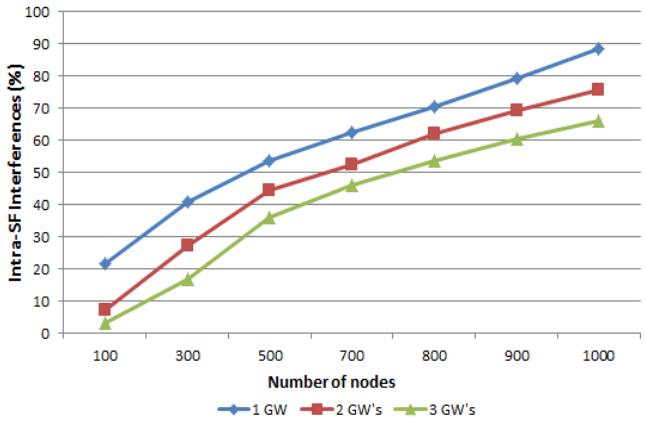

Fig. 14. Intra-SF interferences across three gateway scenarios under varying nodes densitie

VII. Conclusion

In this paper, we investigated the effectiveness of intelligent radio resource allocation in LoRaWAN, focusing on optimizing spreading factor and transmission power allocation through reinforcement learning in a realistic environment. The proposed approach integrates K-means clustering and leverages reinforcement learning to effectively manage gateway placement and autonomously configure transmission parameters based on local signal conditions and environmental feedback. The model explicitly accounts for the capture effect and the imperfect orthogonality of spreading factors, allowing for more realistic and robust adaptation strategies without relying on a centralized ADR mechanism.

Our solution was validated through experimental validation and extensive simulations under various network configurations, demonstrating that the distributed RL-based model outperforms conventional and heuristic RL-based approaches, showing only a small discrepancy with centralized optimal solution in terms of data extraction rate and energy efficiency. Additionally, we highlighted the novelty of our approach by a dedicated analysis of LoRaWAN SFs sensitivity threshold characterization, and evaluated the impact of gateway densification and SF reallocation on network performance. Furthermore, it is noteworthy that the trained learning model performs effectively regardless of the end-device distribution. By sharing the Q-table across all devices, the proposed RL approach can also be extended to mobile network scenarios.

Future work will focus on investigating the feasibility and performance of mobile LoRaWAN networks, including leveraging non-terrestrial infrastructures (e.g, drones) to improve data collection. Addressing these challenges will be a key focus of our upcoming research.

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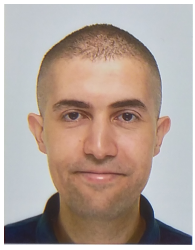

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